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Roll No. | | | | | | | | | | | | | **Total No. of Pages : 03**

Total No. of Questions : 09

B.Tech. (Sem.–2)

MATHEMATICS-II

Subject Code : BTAM-203-18

M.Code : 91959

Date of Examination : 23-01-2023

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION - B & C. have FOUR questions each.**
3. **Attempt any FIVE questions from SECTION B & C carrying EIGHT marks each.**
4. **Select atleast TWO questions from SECTION - B & C.**

SECTION-A

1. Write short notes on :

a) Solve $p = \log(px - y)$

b) Find Integrating factor of $(x^2y^2 + xy + 1)\, ydx + (x^2y^2 - xy + 1)\, xdy = 0$.

c) Find non-ordinary (singular) points of equation $(1-x^2)\dfrac{d^2y}{dx^2} - 2x\dfrac{dy}{dx} + m(m+1)\,y = 0$, m is any real or complex number.

d) Solve : $\sec^2 x \tan y\, dx + \sec^2 y \tan x\, dy = 0$.

e) Solve $\dfrac{dy}{dx} + \dfrac{y}{x} = e^x$. $(x > 0)$

f) For what values of z, the function $w = u + iv$ defined by $z = e^{-v}(\cos u + i \sin u)$ ceases to be analytic.

g) Prove that the function $v = e^{-x}(x \cos y + y \sin y)$ is harmonic.

h) Prove that the image of the straight line $y = mx$ is an equiangular spiral under the transformation $w = e^z$.

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i) Evaluate $\int_c (3z^2 + 2z + 1)\,dz$, where C is the arc of the cycloid $x = a\,(\theta + \sin\theta)$, $y = a\,(1 - \cos\theta)$, between $\theta = 0$ to $\theta = 2\pi$.

j) State Cauchy-Goursat Theorem.

SECTION-B

2. a) Determine for what values of a and b, the differential equation $(y + x^3)\,dx + (ax + by^3)\,dy = 0$ is exact.

b) Solve : $x^2 \dfrac{d^2y}{dx^2} + 2x\dfrac{dy}{dx} - 20y = (x+1)^2$.

3. a) Solve : $x^2\left(\dfrac{dy}{dx}\right)^2 - 2xy\dfrac{dy}{dx} + 2y^2 - x^2 = 0$.

b) Solve : $y'' - 6y' + 9y = \dfrac{e^{3x}}{x^2}$.

4. a) Solve : $y - 2px = \tan^{-1}(xp^2)$.

b) Solve : $x\dfrac{dy}{dx} + y = x^3y^6$.

5. Solve in power series, the differential equation $(1 - x^2)\,y'' - 2xy' + 6y = 0$.

SECTION-C

6. a) Find the analytic region of $f(z) = (x - y)^2 + 2i\,(x + y)$.

b) Evaluate $\int_c \dfrac{z+1}{z^4 - 4z^3 + 4z^2}\,dz$, where C is $|\,z - 2 - i\,| = 2$.

7. a) Prove that $\tan^{-1}\dfrac{y}{x}$ is harmonic.

b) Find the Taylor series to represent the function $\dfrac{z^2 - 1}{(z+2)\,(z+3)}$ in $|\,z\,| < 2$.

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8. a) Find the analytic function $f(z) = u + iv$, whose real part u is $\dfrac{x}{x^2 + y^2}$.

b) Evaluate $\int_c z^2 e^{\frac{1}{z}} dz$, where C is $|z| = 1$.

9. Find the Laurent's series expansion of $f(z) = \dfrac{z+4}{(z+3)\,(z-1)^2}$ in the region $|z - 1| > 4$.

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